Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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Chapter 1

Introduction

Chapter 2

Literature Review

2.0.1 Definition

Biomedical Information Extraction (IE) is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-test articles. This process is usually comprised of two main tasks: Named Entity Recognition (NER), which identifies biomedical entities, and Relation Extraction (RE), which identifies the relationships between those entities [1].

2.0.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with previous noncurrent analyses demonstrating an exponential growth in the number of biomedical publications [2]. This growth is supported by more recent data - a mapping of over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to the level that the volume of biomedical and life-science publications has made it difficult for researchers to keep track of new literature [3]. This is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [4], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

2.0.3 Tasks

2.0.4 Challenges

2.0.5 Traditional Methods

2.0.6 Transition to LLMs

Chapter 3

Methodology

3.0.1 Research Design

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Chapter 4

Results

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Discussion

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Conclusion

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